Reporting delay and calendar clustering in NUFORC public sighting reports

Group name	Dazzling Dataminers
Dataset	`https://www.kaggle.com/datasets/joebeachcapital/ufo-sightings` (file `nuforc_reports.json`)
Code	`https://github.com/HardikGargg/dazzlingDataminers` (notebook and README)

#	Name	Roll no.	Institute email	Contribution
1	Manaswi Singh	2023307	manaswi23307@iiitd.ac.in	loading, before numbers, date and delay cleaning
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3	Divyanshi	2023209	divyanshi23209@iiitd.ac.in	80/20 split, exploratory analysis and figures
4	Hardik Garg	2024228	hardik24228@iiitd.ac.in	hypotheses and Tests 1–4 with alternatives
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1. Dataset

Source: the Kaggle dataset "UFO Sightings" (Arvidsson, 2023). The reports were collected by the National UFO Reporting Center (NUFORC), a voluntary public reporting service, and scraped from its website by Timothy Renner. We use the raw file `nuforc_reports.json`. One row is one public report, not a confirmed event. We do not use the processed CSV in the same download, because its dates and places have already been cleaned.

What is raw about it: the sighting date in `date_time` has a two-digit year; duration is free text (`5 minutes`, `2–3 min`, `ongoing`); country is typed by hand and has 440 spellings; some fields contain the word `null`; 2,044 rows are automatic alerts from sensor stations rather than human reports; and 14 submissions were posted twice.

	Raw file	After encoding
Instances (rows)	141,261	141,247
Attributes (columns)	11	55
Instances $\times$ attributes	1,553,871	7,768,585

Size check: $141{,}247 \times 55 = 7{,}768{,}585 \geq 3{,}000{,}000$, so the requirement is met. The raw file is below the requirement. One-hot encoding of shape (19 groups), country (6 groups) and month (12) takes it past 3M, together with 7 numeric columns and 11 indicator columns from cleaning.

Key features

Feature	Type	Real-world meaning
`day`, `month`, `year`	discrete	date the witness says the sighting happened
`weekday`	discrete	day of the week; Saturday and Sunday count as weekend
`hour`	discrete	hour of the sighting, 0 to 23
`duration_seconds`	continuous	how long the witness says it lasted; the response in Tests 2 and 3
`lag_reported_days`	discrete	days between the sighting and the report being filed
`shape`	nominal	the witness's label for the object, 19 groups; the response in Test 4
`country`	nominal	country of the sighting, 6 groups
`report_link`	nominal	NUFORC's unique identifier for the report

The response in Test 1 is derived: the number of reports on each calendar date.

2. Data preparation

Issue found	What we did	Why, and not the alternative
Two-digit year in `date_time`	took the date from the `Occurred` field of `stats`, which has a four-digit year; `date_time` used only when it was missing (9 rows)	guessing the century could move a 1960s report into the 2060s; the four-digit year needs no guess
11,025 dates typed as a year or a month only	marked as not precise and left out of every day-of-month analysis	NUFORC fills in the 1st for these: 20.0% of them fall on day 1, against 5.0% of precise dates; keeping them would create a false peak
366 rows with no date, 20 with a year before 1900	marked, rows kept	the rows still have a shape and a duration; dropping them would lose that for no gain
Delay below zero	kept 2,759 delays of exactly -1 day; marked 182 delays below -1 day as errors	a delay of -1 day most likely comes from NUFORC recording submissions in its own time zone; larger negative delays cannot be explained
Duration is free text	converted to seconds: ranges take the midpoint, phrases such as `1 hr 30 min` are added; 13,224 texts with no number left missing	durations are very skewed, so filling with a mean or median would invent a length that no witness gave
39 shape labels, partly repeated in two cases; 6,298 missing	lower-cased; missing set to `unknown`; labels with fewer than 1,000 reports merged into `other` (19 groups)	the witness's own label is kept; we do not guess a shape from the text; rare labels would give columns based on very few rows
440 country spellings	mapped to USA, Canada, United Kingdom, Australia, other and unknown	one country appears under many spellings; non-US rows are kept, since a missing state usually means a non-US report
The word `null` in 70 cells; 34 empty texts	set to missing	otherwise `null` would count as a city, a state or a duration
14 submissions posted twice	dropped the second copy	an identical header, down to the second of submission, means one witness counted twice; reports that only share a city and a night are kept
2,044 automatic sensor alerts	marked and left out of all analyses	no person saw anything; 99.6% have no shape and no duration
1,288 reports marked as hoax by NUFORC	marked and kept; every test repeated without them	the hoax label is NUFORC's judgement, not ours
Encoding	one-hot encoded shape, country and month; city not encoded	city has 28,836 names, which would add one column per town

3. What changed after preparation

Measure	Before	After
Rows	141,261	141,247
Attributes	11	39 after cleaning, 55 after encoding
Cells	1,553,871	7,768,585
Sighting date with a four-digit year	none	140,861, of which 129,836 precise
Duration in seconds	none (free text; 7,596 missing)	120,402 (90.1% of non-empty texts)
Shape	39 labels; 6,299 missing	19 groups; none missing
Country	440 spellings; 315 missing	6 groups
Repeated report identifiers	0	0
Submissions posted twice	14	0
Rows that are not human reports	not identified	2,078 marked

Only 14 rows were removed. Every other problem became an indicator column, so each test states which rows it uses. What we lost: 366 rows have no date, 11,025 dates are too imprecise for day-level questions, and 20,845 rows have no usable duration. What we gained: a numeric date, delay and duration, which make every test possible, and a table that meets the size requirement.

4. Basic questions and problem statement

Before making any plot, we split the 141,247 reports at random into an exploration set of 112,998 (80%) and a confirmation set of 28,249 (20%), with seed 42. The split is by report, not by year, so both sets cover the same years. The report identifiers of each set are saved in the repository. The figures below use the exploration set without the non-human rows (111,371 reports). The tests in Section 5 use only the confirmation set.

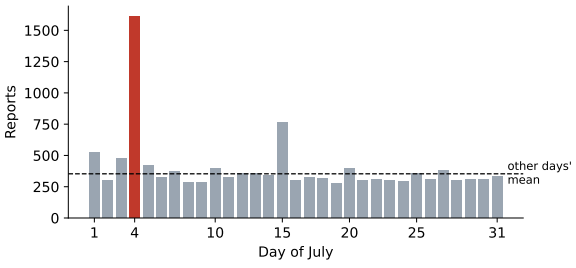


Figure 1: July 4th has 1,613 reports, 4.6 times an average other July day (353). The smaller peaks on the 1st and 15th come from witnesses guessing the day: in every month, 5.0% and 5.7% of precise dates fall on these days, against 3.3% expected.

Basic questions

1. Which state has the most reports? California, with 12.4% of US reports.
2. Which shape is most common? Light, in 20.5% of reports.
3. Do reports cluster at night? Yes: 79.4% fall between 18:00 and 06:00.

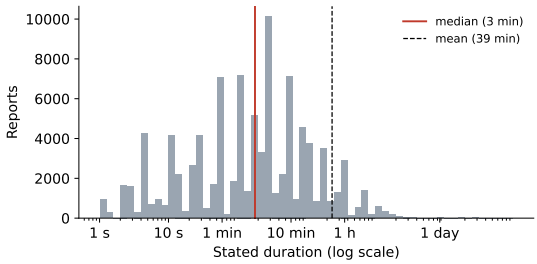


Figure 2: Stated durations are very skewed (median 3 minutes, mean 39 minutes) but nearly symmetric on a log scale, and 52% are round answers such as 5 or 10 minutes.

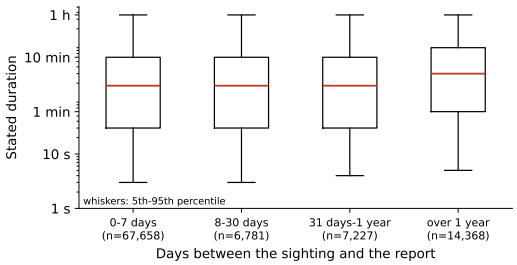


Figure 3: Reports filed within a year of the sighting state the same median duration (3 minutes); only reports filed more than a year later state longer ones (5 minutes).

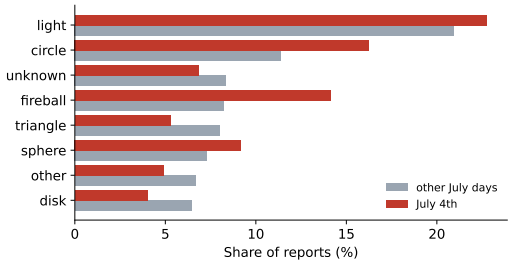


Figure 4: Compared with other July days, July 4th reports name a fireball (14.1% against 8.2%) and a circle (16.2% against 11.4%) more often, while light barely changes.

4. Do reports rise after the late 1990s? Yes: 89.7% of dated reports are from 1998 or later, and the busiest year is 2014.

Our problem statements

P1 (primary). Among July reports, is the share that falls on July 4th larger than the share on other July days, when we compare days of the same kind (weekday or weekend) in the same year?

P2. Do witnesses who file a report long after the sighting state longer durations than those who file soon after it?

How they are novel: the basic questions count reports one field at a time. P1 asks whether the July 4th peak remains in a like-for-like comparison: a plain count could be a weekend effect, since July 4th sometimes falls on a weekend and weekends get more reports than their share of days (32.3% against 28.6%). P2 links two fields that no basic question uses together: the delay between sighting and report, which we derive from two dates, and the stated duration. Either statement could come out either way.

Tests 3 and 4 were added after exploring. Figure 3 suggested that the delay effect appears only after a year, and Figure 4 suggested more fireballs on July 4th. Both are tested on the confirmation set, which played no part in suggesting them.

Both statements concern associations. A July 4th excess would not show that fireworks cause reports, and a link between delay and duration would not show that delay changes memory. We make no causal claim; that would need information on what witnesses actually saw.

5. Statistical inferences

Rows tested: the confirmation set without non-human rows, 27,798 reports. Tests 1 and 4 use July reports with a precise date from the 27 years between 1995 and 2022 that have at least 1,000 reports in the exploration set; days 1 and 15 are left out because guessed dates gather there. Tests 2 and 3 use the 23,953 reports with a duration and a delay not marked as an error. Delay is the date the report was filed minus the date of the sighting. We use the filing date rather than the date NUFORC posted the report, as the proposal planned, because the posting date adds NUFORC's own publishing delay.

Checks: every test is repeated without the 1,288 hoax reports. Because several people may report the same event, Tests 1 and 4 are also repeated with one report per city and night, which sets aside 243 repeats. Tests 2 and 3 are repeated without the 442 reports with a delay of -1 day. All checks are in `outputs/robustness_checks.csv`.

Test 1 (P1): is July 4th over-represented once weekday is held fixed?

a) Hypotheses, written before running the test

H_0	$R \leq 1$
H_1	$R > 1$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	permutation test within each year, 10,000 shuffles
Smallest difference that matters	$R = 1.5$

R is the number of reports on July 4th divided by the number on an ordinary July day of the same year and the same kind (weekday or weekend). The yearly ratios are combined into one weighted average (the Mantel–Haenszel method). The test is one-sided because P1 claims an excess. We chose 1.5 because guessed dates alone raise the 15th to about 1.7 times an ordinary day, so a smaller excess could come from the same kind of error.

b) Other tests we considered

The data are counts of reports per date, and the number of reports differs about tenfold between years. A permutation test fits this directly. In each year we move the July 4th label at random to another day of the same kind, recompute R, and count how often a random day looks as extreme as the real one. It needs no assumption about how the counts are spread.

Also considered	Why we did not use it
Chi-square goodness-of-fit test	assumes that reports are independent and that every July day is equally likely; it ignores year and weekday. Run anyway: $\chi^2 = 1{,}384.7$ (1), $p = 4.5 \times 10^{-303}$.
Exact Poisson test	assumes the variance of the daily counts equals their mean; the variance is 1.37 times the mean within a year and 2.02 times across years. Run anyway: $p = 4.8 \times 10^{-161}$.
Cochran–Mantel–Haenszel test	designed for two-by-two tables of individuals, not counts per date

Both tests that we ran give the same decision, but their p-values are too small to trust, because the counts vary more than these tests assume.

c) Result

Estimate	$R = 5.64$ (15.8 reports per July 4th, 2.69 per comparison day)
95% confidence interval	4.14 to 7.18 (resampling the years)
Test statistic (d.f.)	$R = 5.64$ (none for a permutation test)
p-value	0.0001, the smallest possible with 10,000 shuffles; no shuffle exceeded 2.84
Decision	reject H_0

July 4th has more than five times the reports of a comparable July day. The whole interval lies above 1.5, so the excess matters. Without the hoax reports, $R = 5.63$; with one report per city and night, $R = 5.52$ (95% interval 4.05 to 7.00).

Test 2 (P2): do late filers state longer durations than early filers?

a) Hypotheses, written before running the test

H_0	$P(L > E) \leq 0.5$
H_1	$P(L > E) > 0.5$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	Mann–Whitney U test
Smallest difference that matters	late durations 1.5 times as long

E is a report filed on the same night or within 7 days of the sighting, that is, with a delay of -1 to 7 days (16,826 reports; a delay of -1 day is a same-night report, see Section 2), and L a report filed more than 30 days after it (5,470). $P(L > E)$ is the chance that a random late report states a longer duration than a random early one. We measure the size of the difference as a ratio of geometric means, that is, of averages taken on a log scale. Its interval comes from a Welch t-test on log durations, so it describes the geometric means, not the Mann–Whitney effect; $P(L > E)$ gets its own bootstrap interval. We chose 1.5 because witnesses answer in steps such as 2, 3, 5 and 10 minutes, which are about 1.5 to 2 times apart; a smaller change is finer than what people write.

b) Other tests we considered

There are two independent groups and a very skewed response (skewness 76). The Mann–Whitney test compares ranks, so a few reports of several days cannot dominate it, and it assumes no particular distribution.

Also considered	Why we did not use it
Welch t-test on log duration	valid, since log durations are nearly symmetric (skewness -0.18): $t = 5.82$ (9,545), $p = 3.1 \times 10^{-9}$, the same decision. We kept Mann–Whitney because answers gather at round values and it needs no distribution.
Pooled t-test on log duration	assumes equal spreads, which Levene's test rejects ($p = 0.0004$); $t = 5.72$ (22,294), the same decision
t-test on raw duration	the mean is dominated by a few very long reports: $t = 1.59$ (6,719), $p = 0.056$, the only test that does not reject
Mood's median test	uses only whether each value is above or below the median: $\chi^2 = 27.0$ (1), $p = 2.1 \times 10^{-7}$
Kolmogorov–Smirnov test	detects any difference in shape, not specifically longer durations: $D = 0.042$, $p = 7.8 \times 10^{-7}$

c) Result

Estimate	1.21 times as long (geometric means 171 s and 141 s; medians 210 s and 180 s)
95% confidence interval	1.13 to 1.29 for the ratio (Welch t on logs); 0.516 to 0.533 for $P(L > E)$ (bootstrap)
Test statistic (d.f.)	$U = 48{,}319{,}183$ (none); $P(L > E) = 0.525$
p-value	1.3×10^{-8}
Decision	reject H_0

Late filers state longer durations, but only about 21% longer: for example, 3 minutes becomes about 3.6 minutes. The whole interval lies below 1.5, so the difference is real but too small to matter by our standard. Both intervals point the same way: late durations are slightly longer. Without the hoax reports the ratio is again 1.21; without the -1 day reports it is 1.20 (1.13 to 1.28).

Test 3 (P2): does duration rise steadily with delay, or only for very late reports?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	durations have the same distribution in all four delay bands
$\boldsymbol{H_1}$	at least one band differs
Significance level	$\alpha = 0.05$
Side	no direction for the overall test; each follow-up comparison is one-sided (the later band is longer)
Test	Kruskal–Wallis test, then Mann–Whitney tests between neighbouring bands with Holm's correction
Smallest difference that matters	a ratio of 1.5 between neighbouring bands

The bands are 0–7 days (16,826 reports), 8–30 days (1,657), 31 days to 1 year (1,777) and over 1 year (3,693). Figure 3 suggested that duration changes only after a year; this test checks that on the confirmation set.

b) Other tests we considered

The Kruskal–Wallis test is the rank version of one-way ANOVA. It compares more than two independent groups without assuming a distribution, which suits the skewed durations for the same reason as in Test 2.

Also considered	Why we did not use it
One-way ANOVA on log duration	assumes equal spreads in all bands: $F = 25.6$ (3, 23,949), $p = 1.5 \times 10^{-16}$, the same decision
Alexander–Govern test on log duration	ANOVA without equal spreads, but still assumes normal groups: 81.7 (3), $p = 1.3 \times 10^{-17}$, the same decision
Spearman correlation	looks for a steady trend: $\rho = 0.045$, $t = 6.95$ (23,951), $p = 3.8 \times 10^{-12}$; significant but very weak, because the change is a single step
Jonckheere–Terpstra test	also assumes a steady ordered trend, which Figure 3 does not show

c) Result

Test statistic (d.f.)	$H = 79.4$ (3)
p-value	4.1×10^{-17}
Decision	reject H_0

Step	Ratio	95% interval	p (Holm)
0–7 to 8–30 days	0.98	0.87 to 1.09	1.0
8–30 days to 1 year	0.94	0.82 to 1.09	1.0
1 year to over 1 year	1.49	1.33 to 1.68	5.7×10^{-12}

Duration does not rise steadily with delay. It stays flat for the first year and then rises by about half for reports filed more than a year later. The interval for that step, 1.33 to 1.68, includes 1.5, so we cannot say whether the step reaches the size we set. This also explains Test 2: its late group combines a flat band with this step. The step intervals come from Welch t-tests on log durations and the Holm p-values from Mann–Whitney tests. Without the hoax reports the result is the same (step ratio 1.48), and also without the -1 day reports (1.49).

Test 4 (P1): do July 4th reports describe a fireball more often?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$p_4 - p_o \leq 0$
$\boldsymbol{H_1}$	$p_4 - p_o > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	two-proportion z-test
Smallest difference that matters	3 percentage points

p_4 and p_o are the shares of reports with the shape "fireball" on July 4th (426 reports) and on other July days (2,077), using the same dates as Test 1. The test is one-sided because fireworks would add fireball reports. Three points is more than a third of the share on other days, which is about 9%.

b) Other tests we considered

There are two groups and a yes-or-no response, with at least 60 fireball reports in each group, so the normal approximation behind the z-test holds. The z-test also assumes that reports are independent, which may not hold for reports of the same night; we check this in part (c).

Also considered	Why we did not use it
Chi-square test on the two-by-two table	the same as a two-sided z-test ($\chi^2 = z^2$): 10.73 (1), $p = 0.0011$
Fisher's exact test	exact, but not needed at these sample sizes: odds ratio 1.68, $p = 0.0011$
Chi-square test on all 19 shape groups	tests whether any shape changes, not fireballs in particular, and some expected counts are below 5 (the smallest is 2.4): $\chi^2 = 52.1$ (18), $p = 3.6 \times 10^{-5}$

c) Result

Estimate	$+5.2$ points (14.1% on July 4th, 8.9% on other July days)
95% confidence interval	1.7 to 8.7 points
Test statistic (d.f.)	$z = 3.28$ (none)
p-value	5.3×10^{-4}
Decision	reject H_0

July 4th reports describe a fireball more often, which agrees with witnesses seeing fireworks or lanterns. The interval runs from 1.7 to 8.7 points and so includes 3: the difference is real, but we cannot say that it reaches the size we set. Without the hoax reports the difference is again 5.2 points, and with one report per city and night it is 4.8 points (1.3 to 8.4, $p = 0.0013$).

These checks do not remove the dependence between reports of the same date or year. The fireball share changes a lot between years (it is highest in 2010–2014, the years of sky lanterns), and the z-test ignores this. When we resample whole dates instead of single reports, the 95% interval is -1.5 to 11.0 points; when we resample whole years, it is -0.3 to 10.4 points. Both include zero, so this result is weaker than the z-test suggests and we treat it as suggestive rather than firm.

Correction for running four tests

Running four tests raises the chance of at least one false rejection, so we adjusted the four p-values with Holm's method.

Test	p-value	Holm p-value
1. July 4th excess	0.0001	0.0002
2. Late against early filers	1.3×10^{-8}	3.8×10^{-8}
3. Duration across delay bands	4.1×10^{-17}	1.6×10^{-16}
4. Fireballs on July 4th	5.3×10^{-4}	5.3×10^{-4}

All four tests still reject H_0 after the correction. Tests 1 to 3 hold in every check; Test 4 does not hold once reports from the same date or year are allowed to move together.

6. Limitations

- **Association, not cause.** Our tests show that July 4th has more reports and more fireball reports, and that very late reports state longer durations. They do not show why.
- **Voluntary reporting.** NUFORC receives reports only from people who choose to send them, and 89% come from the USA. Our results describe what people report, not what happens in the sky.
- **Reports are not fully independent.** Several people may report the same fireworks display or the same event. Keeping one report per city and night is only a rough stand-in for one report per event, since one display can be seen from several cities. Test 1 works on counts per date and does not change; Test 4's interval includes zero once whole dates or years are resampled.
- **Dates.** We left out 11,025 imprecise dates, but witnesses who type a full date may still guess the day, as the peaks on the 1st and 15th show. Some people may also round a date to the 4th. NUFORC sometimes misreads compact dates; for example, an entry of `010718` was stored as 7 January 2020.
- **Durations.** The 20,845 rows without a usable duration are not in Tests 2 and 3, and the durations we have are rounded estimates by witnesses.
- **Delays.** We treated delays of exactly -1 day as a time-zone effect. This is likely but cannot be checked from the data; leaving these reports out does not change Tests 2 or 3.
- **Thresholds.** The smallest differences that matter (1.5 times and 3 points) are our own judgement. In Tests 3 and 4 the confidence interval includes the threshold, so a different choice could change whether we call the difference important.